GridRescue AI

GridRescue AI: a grid mapper and advisor that designs a relocatable, theatres-first solar + energy system for a hospital on the move.

BRIEF
12 Relocatable solar for a hospital on the move

TRACK
Clean Energy Solutions

TEAM
Secure AI Grid
Pavan Manjunath · Thomas Prüfer
product, AI & engineering · energy systems & business case

Access · Affordability · Availability "realistic to implement within 6–18 months · adaptable across different global contexts"

HOW THIS PITCH IS SCORED

Impact	25%
Innovation	25%
Sustainability	25%
Feasibility	15%
Scalability	10%

Theatres never dark, on power that moves

THE USE CASE · OUR ANSWER

A federal hospital gets about six hours of grid a day and must move building in five years. GridRescue maps both sites, sizes a theatres-first system that moves with it, and shows the board the outage, the cost and the carbon before anything is bought.

No storage possible (budget, space, site rules)? Grid-first: the plot nearest the strongest grid, a 0 ms UPS for theatres, the generator last; storage as money allows.

Answers the same day a sales or hospital team asks — not after a week-long expert study.

4,992 → 0

theatre-hours a year (40 %) fall when the grid is off today; 0 with the system. The brief's metric, 4 suites, 0 ms switchover

~1,460 → 0

generator switchovers a year (modelled) hitting theatre and ICU equipment — the breakdowns the engineer reports stop at the source

866,678

litres of diesel avoided a year (83 %); 303 kW of life-critical load never off

Judges look for: a large, quantified benefit to the people in the brief, backed by real data and the brief's impact metric.

Design for the move; let open data size it

TODAY

People want to replace diesel with today's energy systems but lack the information to choose among them. Sales and business teams wait days for an expert's MATLAB simulation. Existing grid maps show where the lines are.

OUR APPROACH

GridRescue also reads supply hours, ranks new sites, simulates storage on grid strength and weather, prices it and forecasts the ESG report. Vendor-neutral, it gives that information and makes any supplier's offer comparable.

ROOT CAUSE WE ADDRESS

Root causes 3 and 4: the five-year horizon and the missing financeable design; plus 5 — power quality — with a 0 ms UPS

HOW IT WORKS

1. Map both sites: grid strength, outage hours, sun, tariff
2. Size theatres-first: every storage family is simulated hour by hour against the site's sun, grid hours and load; the hourly simulation picks
3. Price and prove: 5 / 10-year costs, move plan, funders, ESG report

Judges look for: a genuinely new approach that addresses a root cause existing solutions miss, not a copy of what exists.

Paid from the diesel budget, not new capital

COST TO THE USER

₦69.1 M / month

₦0 upfront, versus ₦299.2 M / month of modelled diesel for the protected load. Saves from month one whenever diesel costs more than ₦117.9 M / month.

PAYBACK

1.6 years

10 years: ₦15.9 bn vs ₦37.6 bn on diesel

WHO PAYS

Hospitals: a service fee from the diesel budget, ₦0 upfront (the real diesel bill is partner data still owed). GridRescue: site reports, subscriptions, utility licences; the ranking is never for sale.

HOW IT PAYS FOR ITSELF

Diesel savings repay the system and the year-5 move in 1.6 years. An ESCO with concessional debt owns and moves the kit. Funders: REA Public Sector Solarisation, EU–Enugu Solar for Health, development banks.

WHO KEEPS IT RUNNING

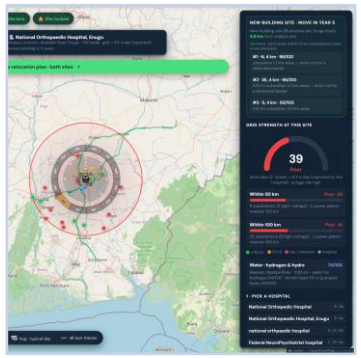
ESCO maintains and monitors remotely; two hospital electricians trained; LFP and PV have local installers; diesel stays the last tier.

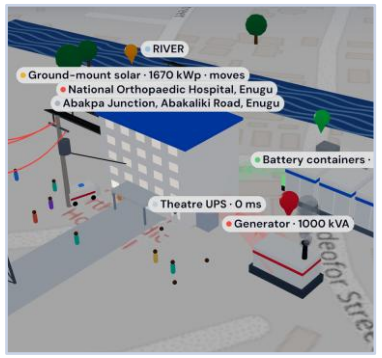
ENVIRONMENTAL BENEFIT

866,678 L of diesel (83 %) and 2,246 t CO_2 avoided a year, net: 2,323 t Scope 1 diesel minus 77 t Scope 2 grid power to charge the batteries. 88 % of the investment reused at the new building.

Judges look for: a solution that pays for itself, is affordable to the lowest-income users, is maintainable locally, and lowers environmental impact.

Works today on open data, for any hospital





EVIDENCE IT WORKS

- 4/4 stress tests pass on Enugu data; any outage up to 24 h survived, whatever hour it starts
  12 h evening outage · 24 h national grid collapse · worst solar month · 18 h outage
- 0 ms UPS for theatres; EERC market; ESCO ownership
- Risk: estimates until the hospital's meter logs replace them

PATH TO SCALE

PILOT Enugu orthopaedic hospital: load audit, EPC quotes, theatres first

EXPAND Federal hospitals on the same tariffs; Dala, Kano; Enugu Solar-for-Health

SCALE Any pin works: sun, weather and grid data are global (NASA POWER, PVGIS, ERA5, OpenStreetMap); a new country needs only its tariff table. Nigeria, Kenya and South Africa are in today; utilities license it.

ALSO ANSWERS Brief 7 (diesel for productive use) · Brief 8 (resilience hubs) · Brief 13 (a city the grid can't protect)

FLEXIBLE next: IoT telemetry, grid planning, cyber-outage tests

OUR ASK **A pilot partner at the Enugu hospital (meter and generator logs, the new plot), the local distribution company to plan grid-first feeders on the same map, and an ESCO or DFI for the first contract.**

Evidence: **Impact** 4,992 at-risk theatre-hours → 0 · **Innovation** built to move · **Sustainability** ₦0 upfront, 2,246 t CO_2 a year · **Feasibility** 4/4 stress tests · **Scalability** any pin